
The effectiveness of electronic games for children with autism

Group 2

Ahdab Ahmed Fairaq 443004727
Reem Saeed Alasmri 443008338
Atheer Mutair Almatani 443006075
Maysam Mohammad Abduljalil 443006214

Supervisor: Dr. Ghader Reda Kurdi

Submission date: 5/2/2024

Abstract

Gamification is a term that has recently been used to depict game applications in areas other than gaming. One of the health conditions where gamification can bring great benefits is to use gamification to help children on the autism spectrum (ASD). The report uses the survey/literature review methodology to collect information related to the impact of electronic games on autistic children in terms of Learning health skills, non-verbal communication, interactive attention, developing executive functions and what are the electronic games that have been developed to help children with autism. Most studies answered the objectives of the report, as they showed a positive impact of electronic games on various skills (health skills, non-verbal skills, interactive attention, and executive functions) in helping children with autism spectrum disorder learn. While the last two studies, achieve the report's objective of reviewing some electronic games developed for autistic children which were the ALTRIRAS game and 'Aalami alssaghir' game, the results were not as expected, for example, the children faced some difficulties with ALTRIRAS game, and also took some time to know the device that was used in 'Aalami alssaghir' game.

Introduction

Autistic children are an essential part of our society and constitute a special group that requires substantial support for their personal, educational, and social progress. Their struggles with both verbal and non-verbal communication significantly impact their ability to integrate with those around them, including relatives and peers. This difficulty may lead to weaknesses in their skills and impede their performance in daily activities. In today's rapidly evolving technological landscape, the play has become an effective tool for educating children. It not only facilitates the exploration of the surrounding world but also improves mental, psychological, and physical health, and fosters the improvement of social and life skills, thereby aiding in their integration as effective members of society.

In this report, the principle of “**gamification**” will be used to determine the extent of the impact and effectiveness of electronic games on the behaviours of children with autism. This report will focus specifically on finding out whether electronic games developed for autistic children can make a difference in their education. Does it positively affect concentration, attention, communication skills, and visual and auditory processing?

The following pages set out the purpose of the report in detail and answer how effective electronic games are for autistic children in four points: learning health skills, non-verbal skills, interactive attention and the development of executive functions electronic games designed for children with Autism Spectrum Disorder (**ASD**) were developed

The report sections:

Section 0.1: Background

define the key definitions

Section 0.2: Related Works

present the related studies

Section 0.3: Methodology and procedure

the methodology that was used

Section 0.4: Results and discussion

results and discussion on the important points

Section 0.5: Conclusion and recommendations

summary of the report and recommendations for future work

Section 0.6: references

present where everything written has been, taken from

0.1 Background

Autism spectrum disorder (ASD):

Autism spectrum disorder (ASD) is a neurodevelopmental disorder that typically occurs in early childhood. It encompasses a wide range of symptoms and levels of impairment, hence the term “spectrum.” In addition to challenges with social interaction, communication, and repetitive behaviours, individuals with ASD may also have sensory sensitivities. These sensi-

tivities can manifest themselves in a variety of ways, such as increased or decreased responses to sensory stimuli such as light, sounds, textures, and tastes.

Gamification:

Gamification integrates game design elements and principles into non-gaming environments to improve user engagement, motivation, and behaviour. By integrating elements such as rules, goals, competition, rewards, and progression systems into traditionally non-gaming environments, gamification aims to make tasks, activities, or processes more enjoyable and engaging. One of the key aspects of gamification is the use of intrinsic elements and extrinsic motivators to drive behaviour change. Intrinsic motivators tap into the individual's inner desires, such as the satisfaction of overcoming a challenge, while extrinsic motivators involve concrete rewards or recognition. By leveraging these motivators, gamification can encourage users to actively participate in activities, adopt new behaviours, or achieve specific goals.

0.2 Related Works

- **Ahmed Al-Hayek's study in 2018**, where the study aimed to achieve two basic goals, the first of which was to evaluate the impact of a training program dedicated to developing healthcare skills among special education specialists working in autism centres. The second goal was to try to identify the relationship between the impact of the program and the variety of educational achievements.

The results of this study also showed that there were statistically significant differences between the experimental and control groups on the health skills scale, which indicates the impact of the training program, and the absence of statistically significant differences between the experimental and control groups on the health skills scale concerning academic achievement.

- **Asmaa Al-Karim's study in 2023**, aimed to identify and discover the effectiveness of a play-based program aimed at developing non-verbal communication that autistic children suffer from. The research concluded that there are statistically significant differences between the experimental and control groups on the scale of non-verbal communication skills.

By presenting previous studies related to the subject of this study and reviewing the results of these studies, it is clear that there are few studies regarding the development of various skills of autistic children (health skills, non-verbal communication skills, etc.) through modern technical methods, as both studies included Trying to develop these skills through regular training programs. This report also distinguishes itself from other previous reports mentioned in that it specializes in showing the impact of one of the most widespread and available games at present, namely electronic games, on several different skills among children with autism, which affects their daily lives in a significant way. The benefit of this report is also evident in the lack of studies related to the subject of this study in particular.

0.3 Methodology and procedure

The report uses the survey/literature review, methodology, and the gathered data will be analysed through the following, two reporters will focus on searching for answers to the aims and the questions of the report using the references, while the other two will verify the collected information. Ensuring the accuracy of formulating the answers of each aim. This methodology was selected and perfect for this report because of the lack of time, and it's difficult to read many reports in this short period, and Some resources were not available for a certain amount of money, also to provide comprehensive previous studies that cover all the objectives of this report and Expanding knowledge regarding the latest research findings on this topic. This report uses the following groups of search keywords: Keywords related to ASDs (autism spectrum disorders): autism - autistic child - autistic children

Keywords related to **gamification**: **gamification**

Keywords related to skills: **health skills**

Keywords related to communication: **Non-verbal communication**

Keywords related to electronic games: **educational Electronic games, Electronic games**

Keywords related to emotions: **emotions**

Search queries include 'electronic games for autistic children' - 'Gamification of autistic children' - 'The effectiveness of electronic games on autistic children' - 'The effect of electronic games on autistic children'. the data that are presented in this report were gathered from 6 studies that are from the Google Scholar database¹ and a survey from a graduation project in the 'Enjaz' exhibition of (Umm al-Qura University)², and the Saudi Digital Library (SDL) database³. The data is extracted by first picking studies that have the closest title to this report's title or the title related to one of the objectives, then going through the abstract of each study and checking the objectives, the study question if found, and the results. To ensure related and suitable sources of information. After that, the detailed results of each report were checked and summarized. This report uses a table to analyze the extracted data(the results of the studies) in terms of the authors, focus (what did these studies focus on?), the target group, the results, and the evaluation method (how the studies evaluate the data?) and the methodology of each study, providing with this table, an explanation of each study results.

0.4 Results an discussion

all the tables listed under describe studies that most of them focused on the cognitive and auditory aspects of children with autism, while most studies target groups where between (4-12), and the experimental methodology was used in most of the studies also, and the evaluation method that was used for more than 50 per cent of the studies was a set of questions to be answered by

¹<https://scholar.google.com/>

²<https://publuu.com/flip-book/388065/880394>

³<https://sdl.edu.sa/SDLPortal/en/Publishers.aspx>

children with autism.

In the study of **Al-Qahtani, Khalid bin Nasser bin Mudhakir (2016)**., which is presented in Table 1, there is a positive effect of the adaptive learning environment based on the use of educational games on developing non-verbal communication skills for the benefit of the experimental group in the post-application. The results of the sequential application, one month after the post-application of the learning environment, also indicated that the children had not forgotten their non-verbal communication skills and that there had been no relapse in these skills.

In the study of **Jablawi, Raneem Khairaat (2021)**, which is presented in Table 2, the experimental group outperformed the control group in the post-test, which indicates that the health skills of autistic children can be developed and developed through educational electronic games. The study also showed that the development of these skills occurred equally in males and females, without any difference between the sexes. The delayed post-test showed that these health skills have become behavioural habits for autistic children. In the study of **Muhammad, Hind Adel, Hassanein, Iman Salah al-Din Saleh, Ahmed, and Muhammad Youssef(2022)**, which is presented in Table 3, the results showed that there is a statistical hypothesis between the average bicycles of the two experimental groups and improving reactive attention in autistic children in the two experimental groups

The results also showed an increase in the percentage of response rates in favour of people with mild autism in the electronic test, where the game was presented attractively and easily. They took into account individual differences which led to improving the attention skills of each of the two groups. There is also a statistical difference between the average scores of the two experimental groups in favour of children with simple autism, as the values ranged between (0.56:0.85), which indicates the positive educational electronic games in improving interactive attention skills among children with simple and medium autism.

In the study of **Musa, Amani Youssef El-Sayed Mohammed (2021)**, which is presented in Table 4, the results of the study indicated the effectiveness of electronic educational games in improving some of the executive function skills of autistic children, as an appropriate educational environment was created that enables them to benefit from electronic educational games. This program contained training activities accompanied by sound and repeating the sound several times, in addition to accompanying it with pictures indicating the word heard. Which led to an increase in awareness of children with autism.

In the study of **Alqurashi, Alzahrani, Alsuwayh and Alobaidi (2023)**, which is presented in Table 5, they have designed two stages for this game: the Kitchen stage and the Classroom stage.

In the Kitchen stage, the mother character will appear and teach the child about sharp tools that that he is not supposed to play with. The mother will show two tools from the kitchen and ask the child which one is dangerous and which one they should not play with or hold. The child will answer by using the control stick.

In the Classroom stage, the teacher will ask the students if they want to answer a question,

and the children must raise the hands. The child can interact with the teacher and wave their hand using the control stick. This action will enable the child to answer the question. The question will appear on the board, and the child can provide their answer.

In the study of **Almeida et al (2019)**, which is presented in Table 6, the results were presented through the following elements:

1. Expert evaluation and usability testing: The experts considered ALTRIRAS to be certified in all quality metrics evaluated (graphics, sounds, controls, fun, and animation). According to them, the method meets psychological and entertainment requirements, and audio stimuli are able to enhance learning in children.

2. Usability and effectiveness tests with non-disabled children: Regarding non-disabled children, when they took the test, the experts considered that most of the children (57 per cent) found the game easy for them, while others (43 per cent) found that the game was the best according to the tests they conducted, and the results were evaluated using the SUS scale, and they were more positive about evaluating the game's ease of use.

3. Testing effectiveness with children with autism spectrum disorder: The autistic children showed noticeable difficulty in answering the questions in the game, according to the Wilcoxon applied statistical test, there was no significant difference in the number of their correct answers before and after. The psychiatrist suggested that this game be included in their daily routine to show the difference in the results because they have more difficulty in learning, than normal children.

Of the reviewed studies, the first four studies answered the report objectives, since they showed a positive impact of electronic games over different skills (health skills, non-verbal skills, interactive attention, executive functions), in helping children with ASD to learn and develop these skills to be a part of their daily routine. for example the study (jablawi,2021), the study adopted the use of colourful and fun electronic games, which led to the experimental group outperforming the control group in their post-application of the test with a high effect value of 0.43.

While the last two studies, achieve the report's objective of reviewing some electronic games developed for autistic children which were the ALTRIRAS game and 'Aalami alssaghir' game, the results were not as expected, for example, the children faced some difficulties with ALTRIRAS game, and also took some time to know the device that was used in 'Aalami alssaghir' game.

Authors	Al-Qahtani, Khalid bin Nasser bin Mudhakir . (2016).
Focus	Designing an adaptive learning environment based on the use of free electronic games via the web (TAC, Autism iHelp) and discovering their impact on developing participation and non-verbal communication skills (Eye contact, listening and paying attention, expressing with gestures, hand waving and imitation, in addition to understanding and showing facial expressions) for children with <u>Autism Spectrum Disorder</u> (ASD).
Target group	17 children with autism disorder (ASD) aged (4-8) years divided into two groups: a group of 9 children (experimental) and a group of 8 children (control).
Methodology	Descriptive analytical method, quasi-experimental method.
Evaluation method	A note card to estimate nonverbal communication skills (he achieves the skill and gets a score of one, he does not achieve the skill and gets a score of zero) consisting of 20 skills.
Results	1. There are no statistically significant differences between the average ranks of the children in the experimental and control groups in the pre-measurement of the observation card. 2. There are also statistically significant differences between the autistic children in the experimental and control groups in the post-application of the observation card, which was in favor of the experimental group. 3. There are also statistically significant differences. Among the members of the experimental group in the pre-and post-applications in favor of the post-application.

Figure 1: **Table 1**

Authors	Jablari, raneem khairaat (2021)
Focus	Verifying the effectiveness of educational electronic games (Babybus) in developing some health skills (using the bathroom, washing clothes, washing hands before eating, eating healthy food, washing the face and teeth before sleeping) among a sample of children with <u>Autism Spectrum Disorder</u> (ASD).
Target group	16 children with autism disorder (ASD) aged (5-6) years divided into two groups: group of 8 children (experimental) and a group of 8 children (control).
Methodology	quasi-experimental method.
Evaluation method	The list of health skills consisted of 26 questions regarding these skills, where the autistic child gets a score of one if he gives a correct answer and a zero if he gives an incorrect answer.
Results	1. There are statistically significant differences between the average grade ranks of autistic children in the control and experimental groups in the post-application, which is in favor of the experimental group, as the value of the effect was high 0.43. 2. and there are statistically significant differences between the average grades of the children in the experimental group in the pre-application and the post-application, in favor of the post-application, where it reached the impact rate is high 0.44. 3. and there are no statistically significant differences between the average ranks between males and females among the children of the experimental group in the post-application of the health skills test. 4. There are also no statistically significant differences between the average ranks of the children's scores in the experimental group in the direct post-application and the follow-up post-application.

Figure 2: **Table 2**

Authors	Muhammad, Hind Adel Hassanein, Iman Salah al-Din Saleh, Ahmed, Muhammad Youssef (2022).
Focus	Observing the level of interactive attention and measuring the impact of educational electronic games on the development of attention skills in mild and moderate autistic children after playing a simple educational electronic game with drag-and-drop skills.
Target group	10 children with autism disorder (ASD) Two groups of autistic children: 5 children with mild autism 5 children with moderate autism
Methodology	Descriptive analytical method, experimental method
Evaluation method	Two measuring tools: Electronic test: 5 screens Screen 1 & 2 - Shape matching axis, matching the shape in its correct place. Screen 3 & 4 - Size matching axis, choosing and matching the appropriate size to its correct place. Screen 5 - Partial completion of shapes axis, completing the existing shape with its correct part from among several options. Note card for attention skills: 9 sentences that measure the ability to interact with the game and. Responding with screens to achieve the goals in each sentence.
Results	1. There are statistically significant differences between moderately autistic and mildly autistic people in the pre-measurement test The research is for the benefit of people with autism, and it also indicates the ability of the electronic test to distinguish between different groups. 2. There are statistically significant differences between moderately autistic and mildly autistic people in the post-measurement test The research is for the benefit of people with autism, and it also indicates the ability of the electronic test to distinguish between different groups. 3. There are statistically significant differences between those with moderate autism and those with mild autism in the pre-measurement using the card under study, in favor of those with mild autism, and it also indicates the ability of the electronic card to distinguish between different groups. 4. There are statistically significant differences between those with moderate autism and those with mild autism in the post-measurement using the card under study, in favor of those with mild autism, and it also indicates the ability of the electronic card to distinguish between different groups.

Figure 3: Table 3

Authors	Musa, Amani Youssef El-Sayed Mohammed (2021)
Focus	effectiveness of a training program using electronic educational games in improving some executive function skills (planning, cognitive flexibility, and working memory) among children with autism disorder.
Target group	10 children with autism disorder ages (9-12).
Methodology	Experimental Methodology
Evaluation method	It was tested at Stanford University and is a test of ten sub-tests divided into two main areas (verbal and non-verbal). Each area contains five sub-tests, and each sub-test consists of a set of <u>mini-tests</u> of varying difficulty (starting from easiest to most difficult).
Results	It was observed that the abilities of the child with autism improved in terms of attention and discrimination.

Figure 4: **Table 4**

Authors	<u>Alqurashi</u> , <u>Alzahrani</u> , <u>Alsuwayh</u> and <u>Alobaidi</u> (2023)
Focus	Designing a game using virtual reality technology to improve the basic skills of autistic children and enhance education methods for these children in the Kingdom of Saudi Arabia.
Target group	A group of 10 children in the training session
Methodology	Experimental Methodology
Evaluation method	Bringing the children to the training session and teaching them about the virtual reality device and how to play with it.
Results	The children took some time to get to know the device because it is not included in their daily <u>routine</u> and it is considered a new device for them.

Figure 5: **Table 5**

Authors	Almeida et al (2019)
Focus	Designing a computer game to train Autistic children to recognize basic human emotions: Anger, sadness, joy, surprise. And associate the emotions with their corresponding facial expressions.
Target group	10 children with ASD
Methodology	Experimental Mehodology
Evaluation method	To gather data on the children's performance and interaction with the computer game, they were brought to the training facility, where they were given the opportunity to play the game. Throughout the session, their results and interactions with the game were closely monitored and recorded for analysis.
Results	The results obtained from the game development process, as well as the effectiveness testing conducted by the subjects, are presented in the following elements: 1. ALTRIRAS game: The scientific article included pictures of the game's interfaces, which were reviewed at the end of the table. 2. Expert evaluation and usability testing: The game underwent evaluation by experts who utilized the System Usability Scale (SUS) to assess its quality components. 3. Usability and effectiveness tests with non-disabled children: The game was also tested with non-disabled children, and they demonstrated more positive results in terms of evaluating the game's ease of use. 4. Testing effectiveness with children with autism spectrum disorder: When the game was tested with children with autism spectrum disorder, they encountered significant difficulties.

Figure 6: Table 6



Figure 7: Home interface ALTRIRAS



Figure 8: Opening of the box and receiving a facial expression ALTRIRAS



Figure 9: Teaching facial expressions, anger ALTRIRAS



Figure 10: Training emotions ALTRIRAS



Figure 11: great if the kid wins ALTRIRAS



Figure 12: try again if the kid loses ALTRIRAS

0.5 Conclusion and recommendations

The learning environment plays a major role in the autistic child's response to treatment, and modern methods such as electronic educational games have a positive effect, and the child's response is faster and his performance improves significantly. One of the obstacles we faced was the lack of sources and the lack of time was a major factor, which in turn reduced the number of references used in this report. To create the appropriate environment for this, it is necessary to make specialists apply modern treatment methods and develop training courses.

Based on the findings, the following can be recommended:

1. Conducting training workshops for professionals to learn modern methods.
2. The need to diversify the activities offered to them.
3. Involvement of the family in the prepared programs.
4. Providing financial payment options Attention to studies and research on the Category of autistic children.

0.6 References

- Nasser Mudhakhir Al-Qahtani, Kh., & Khaled. (2016). Designing an adaptive learning environment based on the use of educational electronic games via the web to develop participation and non-verbal communication skills in children with autism disorder. Educational Technology: Studies and Research Series, 26 (Issue Four, Part One), 255-297.
- Jablawi, R. K. (2021). The effectiveness of educational electronic games in developing some health skills among autistic children - a quasi-experimental study in Lattakia Governorate. Tishreen University Journal-Arts and Humanities Sciences Series, 43(5).
- Musa, A. Y. a. M., & Amani Youssef Al-Sayed Muhammad. (2021). The effectiveness of a training program using electronic educational games in developing executive functions for children with autism disorder. Educational Sciences: A quarterly peer-reviewed journal, 29(3), 450-498.
- Almeida, L. M., Silva, D. P. D., Theodório, D. P., Silva, W. W., Rodrigues, S. C. M., Scardovelli, T. A., ... & Bissaco, M. A. S. (2019). ALTRIRAS: A computer game for training children with autism spectrum disorder in the recognition of basic emotions. International Journal of Computer Games Technology, .2019
- Muhammad, Hind Adel, Hassanein, Iman Salah al-Din Saleh, and Ahmed, Muhammad Youssef. (2022). Educational electronic games and their impact on developing interactive attention skills for children with disabilities Mild and moderate autism. Journal of Research in the Fields of Specific Education, No. 39 48. - 1, Retrieved from <http://search.mandumah.com/Record1>
- Ahmed Al-Hayek. (2018). The impact of a training program on developing health care skills for autistic children among those who care for them in autism centers and its relationship

to some variables in a Jordanian sample. *International Journal of Educational Psychological Studies (EPS)*, 3(2).

- Saber Hamed Abdel Karim, D. (2023). The effectiveness of a play-based training program in developing nonverbal communication in autistic children. *Studies in Psychological and Educational Counseling*, 6(2), 88-130.

- Alqurashi, Alzahrani, Alsuwayh and Alobaidi (2023). Alami alssaghir[Poster]. Retrieved from <https://publuu.com/flip-book/388065/880394>